

Reflection

you are acting as an senior coding engineer my background a senior college student acting as project manager. my goal is to design a project i can present in class. Constraints: i cannot write code manually, i must rely on ai tools, the project must be feasible in 2 weeks. please 1. break the project into steps 2 tell me what i should ask ai at each step 3. warn me about common mistakes this is the idea Mini App Concept Image-Aware Caption Generator with Vibe Control One-liner: Upload a photo, choose a vibe, get captions that actually fit the image — not generic AI fluff. This is not “just another caption generator.” The differentiation is image understanding + vibe alignment. The Real-World Problem (Why This Exists) Most caption tools: Ignore the image Produce generic, cringe, or off-tone captions Don’t respect why someone is posting Users struggle with: Matching caption tone to the image mood Knowing how to say something, not what Avoiding captions that hurt engagement Your app solves tone + context mismatch. Target Users (Start Narrow) Primary: Instagram users (personal + small creators) Small business owners posting products Casual influencers (1k–50k followers) Secondary: Discord users (image posts) LinkedIn users (professional vibe later) Core User Flow (Very Important) Step-by-step UX User uploads an image User selects a vibe (Optional) User adds: Platform (Instagram / Discord) Length (short / medium / long) AI analyzes the image AI generates 5 captions User copies or regenerates This flow should take under 30 seconds. Vibe System (Your Key Differentiator) Start with clear, human vibes: Vibe examples Funny 😄 Aesthetic ✨ Professional 📁 Emotional ❤️ Confident 🔥 Casual 🗨 Minimal 🧘 Each vibe changes: Word choice Emoji usage Sentence length Punctuation style Think of vibes as style presets, not moods.

Additional constraints::

- keeping everything in a single file
- reading the API key from an environment variable
- simplifying the UI to focus on one clear task

The AI helped debug many issues I didn’t anticipate, environment variables, Streamlit execution, API errors, and quota limits. It also suggested using Streamlit for quick UI, which I had not originally planned.

mini-app is an image caption generator. A user uploads an image, and the app generates a caption suitable for social media.

The problem it solves is reducing the friction of coming up with captions that match the image content.

satisfied with is that the app is runnable, self-contained, and demonstrates real AI interaction rather than static output.

how much iteration was needed. The AI was okay at fixing errors and explaining deployment issues, but it still required me to think carefully about constraints and requirements.

The AI felt more like a collaborator more so than a tool — especially during debugging and clarification. It helped move the project forward, but I still had to double check what was done

Note I hit quota limit for this app project so couldn't test in detail.

The golden rule allowed me to step through the AI feedback and communicate more effectively for a better final product. Knowing common errors, mentioning to AI to prompt this was super important.